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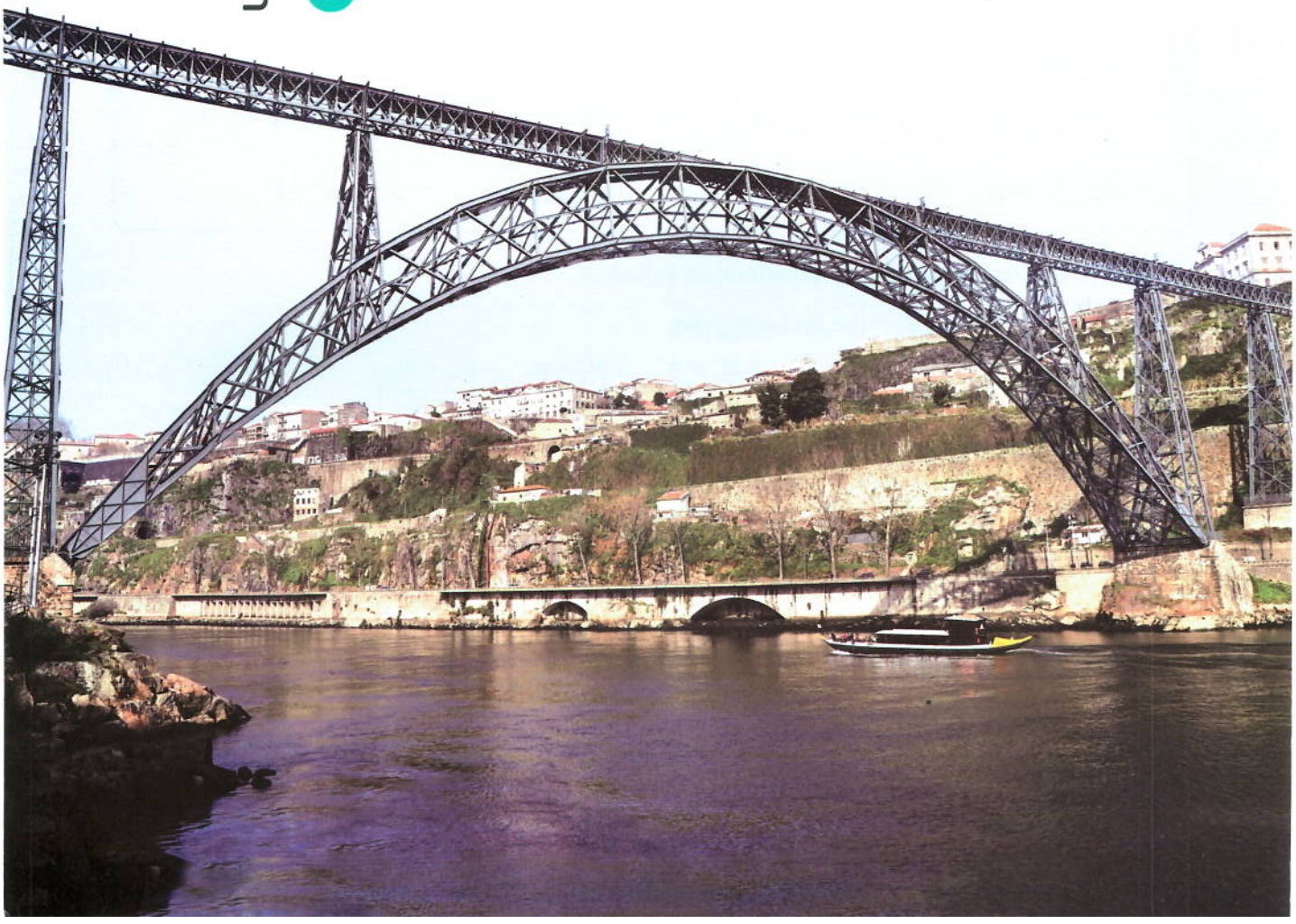
Differential & Integral Calculus

Revision on what have been studied before.

Unit 1 Differentiation and its applications.

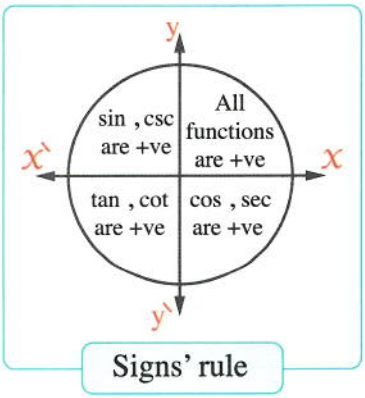
Unit 2 Behavior of the function and curve sketching.

Unit 3 The definite integral and its applications.



Revision on what have been studied before

Important rules of trigonometry



Signs' rule

- **The basic trigonometric identities**

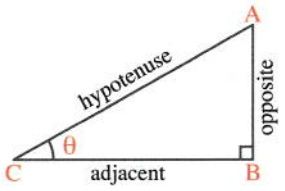
For any angle of measure (θ), then :

1. $\cos^2\theta + \sin^2\theta = 1$ **2.** $1 + \tan^2\theta = \sec^2\theta$

3. $1 + \cot^2\theta = \csc^2\theta$ **4.** $\tan\theta = \frac{\sin\theta}{\cos\theta}$, $\cot\theta = \frac{\cos\theta}{\sin\theta}$

5. $\sin\theta \csc\theta = 1$, $\cos\theta \sec\theta = 1$, $\tan\theta \cot\theta = 1$

- **Remember the following relations :**

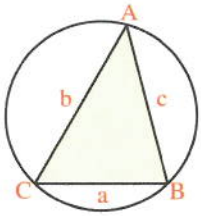


1. $\sin\theta = \frac{\text{opp.}}{\text{hyp.}} = \frac{AB}{AC}$ **2.** $\cos\theta = \frac{\text{adj.}}{\text{hyp.}} = \frac{BC}{AC}$

3. $\tan\theta = \frac{\text{opp.}}{\text{adj.}} = \frac{AB}{BC}$

- **The sine rule :**

In any Δ ABC : $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2r$



where r is the radius length of the circumcircle of the Δ ABC

- **The cosine rule :**

In any Δ ABC : $a^2 = b^2 + c^2 - 2bc\cos A$, $b^2 = a^2 + c^2 - 2ac\cos B$, $c^2 = a^2 + b^2 - 2ab\cos C$

this rule is used if : the lengths of two sides in Δ ABC and measure of the included angle between them are given.

From the previous we get :

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} \quad , \quad \cos B = \frac{a^2 + c^2 - b^2}{2ac} \quad , \quad \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

this rule is used if : the lengths of the sides of Δ ABC or the ratio among these lengths are given.

- Area of $\Delta ABC = \frac{1}{2}$ product of lengths of any two sides × sine the included angle

$$= \frac{1}{2} a\, b \sin C = \frac{1}{2} a\, c \sin B = \frac{1}{2} b\, c \sin A$$

- If $s = \frac{a+b+c}{2}$, then the area of $\Delta ABC = \sqrt{s(s-a)(s-b)(s-c)}$
- Trigonometric function of the sum and difference between the measures of two angles :

1. $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$ **2.** $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$

3. $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$

- Trigonometric functions of double of the measure of an angle :

1. $\sin 2A = 2 \sin A \cos A$

2. $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$

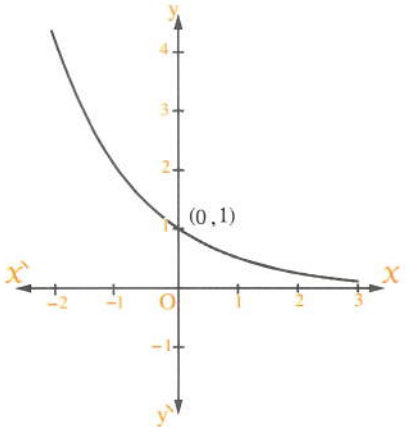
3. $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$

1 Exponential function :

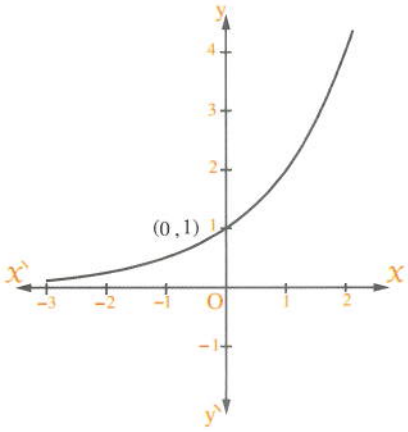
* If $a \in \mathbb{R}^+ - \{1\}$, then the function $f : \mathbb{R} \longrightarrow \mathbb{R}^+$ where $f(x) = a^x$ is called exponential function whose base (a) and it is one - to - one function its domain = $\mathbb{R}$, its range = $\mathbb{R}^+$ and its curve passes through the point (0 , 1)

* The graphical representation of the exponential function $f : f(x) = a^x$ is one of the two following shapes , according to the value of a :

$0 < a < 1$



$a > 1$



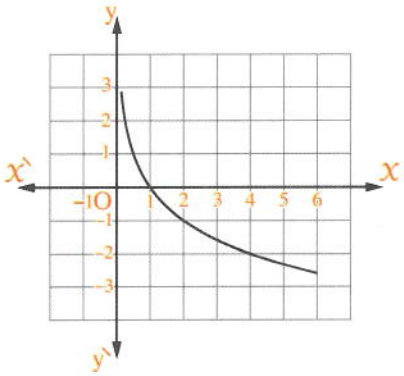
* The curve $y = a^{x+b} + c$ is represented by the same curve of $y = a^x$ by translation $(-b , c)$

2 Logarithmic function :

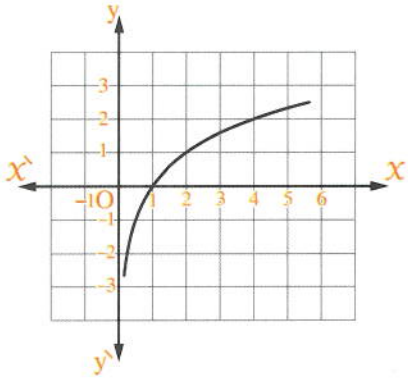
* If $a \in \mathbb{R}^+ - \{1\}$, then the function $f : \mathbb{R}^+ \longrightarrow \mathbb{R}$ where $f(x) = \log_a x$ is called logarithmic function whose base (a) and it is one - to - one function its domain $\mathbb{R}^+$, its range $\mathbb{R}$ and its curve passes through (1 , 0)

* The graphical representation of the logarithmic function $f : f(x) = \log_a x$ is one of the following according to the value of a :

if $0 < a < 1$



if $a > 1$



* The curve $y = \log_a (x + b) + c$ is represented by the same curve of $y = \log_a x$ by translation $(-b , c)$

* The common logarithm is a logarithm whose base is 10

⊙ Some properties of logarithm if $x \in \mathbb{R}^+$, $y \in \mathbb{R}^+$, $n \in \mathbb{R}$ such that :

the base $a \in \mathbb{R}^+ - \{1\}$ then :

1. $\log_a a = 1$

2. $\log_a 1 = 0$

3. $\log_a x^n = n \log_a x$

4. $\log_a x y = \log_a x + \log_a y$

5. $\log_a \frac{x}{y} = \log_a x - \log_a y$

6. $\log_y x = \frac{\log_a x}{\log_a y}$

7. $\log_y x = \frac{1}{\log_x y}$

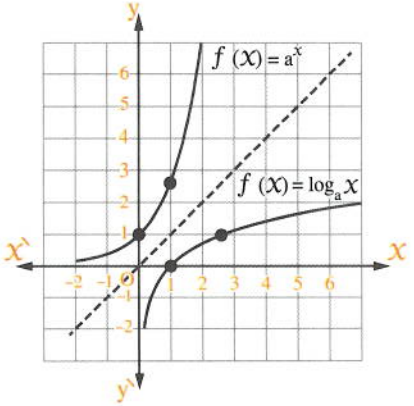
8. $a^{\log_a x} = x$

3 The relation between the exponential function and the logarithmic function :

⊙ If $a \in \mathbb{R}^+ - \{1\}$ $x \in \mathbb{R}^+$, $y \in \mathbb{R}$

then $y = \log_a x \Leftrightarrow x = a^y$

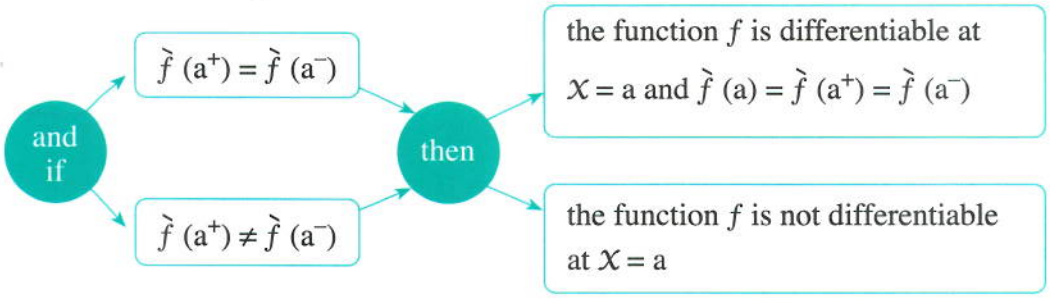
⊙ The logarithmic function is inverse function of the exponential function



Continuity and differentiability

1. We can say that the function f is differentiable at $\mathcal{X} = a$ (where $a \in$ Domain of f) if and only if $\grave{f}(a)$ is exist where $\grave{f}(a) = \lim\limits_{h \to 0} \dfrac{f(a+h) - f(a)}{h}$

2. If $a \in$ Domain of f :



3. If the function f does not continuous at a certain point then it is not differentiable at this point.

4. (If the function f is differentiable at $\mathcal{X} = a$, then it should be continuous at $\mathcal{X} = a$)

But the converse is not necessary ture **i.e.** (If the function is continuous at $\mathcal{X} = a$, then it is not necessary to be differentiable at this point)

5. If derivative of function f at each point belong to the interval $]c , d[$ is exist then we can say that the function f is differentiable on this interval.

6. Any polynomial function is differentiable on $\mathbb{R}$

7. $\grave{f}(a)$ "the 1st derivative of the function f at the point $\big(a , f(a)\big)$"

= rate of change of f at the point $\big(a , f(a)\big)$

$= \lim\limits_{h \to 0} \dfrac{f(a+h) - f(a)}{h}$ (under condition of : the limit is exist)

= slope of the tangent of the curve of the function f at the point $\big(a , f(a)\big)$

Rules of differentiation

1. If $f(\mathcal{X}) = a$ (where a is constant) , then $\grave{f}(\mathcal{X}) =$ zero

2. If $y = [a\, f(\mathcal{X})]$ (where a is constant) , then $\grave{y} = a\, \grave{f}(\mathcal{X})$

3. If $y = \mathcal{X}^n$ (where $n \in \mathbb{R}$) , then $\grave{y} = n\, \mathcal{X}^{n-1}$

4. If $y = f(\mathcal{X}) \pm g(\mathcal{X})$, then $\grave{y} = \grave{f}(\mathcal{X}) \pm \grave{g}(\mathcal{X})$

5. If $y = f(x) \times g(x)$, then $\grave{y} = f(x) \times \grave{g}(x) + g(x) \times \grave{f}(x)$

i.e. $\frac{dy}{dx}$ = the first × derivative of the second + the second × derivative of the first.

6. If $y = \frac{f(x)}{g(x)}$ where $g(x) \neq 0$, then $\frac{dy}{dx} = \frac{g(x) \times \grave{f}(x) - f(x) \times \grave{g}(x)}{(g(x))^2}$

i.e. $\frac{dy}{dx} = \frac{\text{(denominator} \times \text{derivative of numerator)} - \text{(numerator} \times \text{derivative of denominator)}}{(\text{denominator})^2}$

7. Chain rule : $\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$

8. If $y = [f(x)]^n$, then $\frac{dy}{dx} = n[f(x)]^{n-1} \times \grave{f}(x)$

i.e. $\frac{dy}{dx} = n\,(\text{bracket})^{n-1} \times$ derivative of what is inside the bracket.

9. Differentiation of the trigonometric function (where x in radian)

* If $y = \sin x$, then $\frac{dy}{dx} = \cos x$
* If $y = \cos x$, then $\frac{dy}{dx} = -\sin x$
* If $y = \tan x$, then $\frac{dy}{dx} = \sec^2 x$
* If $y = \cot x$, then $\frac{dy}{dx} = -\csc^2 x$
* If $y = \sec x$, then $\frac{dy}{dx} = \sec x \tan x$
* If $y = \csc x$, then $\frac{dy}{dx} = -\csc x . \cot x$

Generally

* let : $y = \sin f(x)$, then $\frac{dy}{dx} = \grave{f}(x) . \cos f(x)$
* let : $y = \cos f(x)$, then $\frac{dy}{dx} = -\grave{f}(x) . \sin f(x)$
* let : $y = \tan f(x)$, then $\frac{dy}{dx} = \grave{f}(x) . \sec^2 f(x)$
* If $y = \cot f(x)$, then $\frac{dy}{dx} = -\grave{f}(x) . \csc^2 f(x)$
* If $y = \sec f(x)$, then $\frac{dy}{dx} = \grave{f}(x) \sec f(x) . \tan f(x)$
* If $y = \csc f(x)$, then $\frac{dy}{dx} = -\grave{f}(x) \csc f(x) . \cot f(x)$

Geometric applications

Slope of the straight line :

1. Slope of the straight line which its equation :

$a\mathcal{X} + by + c = 0$ is $\frac{-\text{coefficient of } \mathcal{X}}{\text{coefficient of y}} = \frac{-a}{b}$

2. Slope of the straight line which passes through the two points $(\mathcal{X}_1 , y_1) , (\mathcal{X}_2 , y_2)$

equals $\frac{y_2 - y_1}{\mathcal{X}_2 - \mathcal{X}_1}$

3. Slope of the straight line = $\tan\theta$

where θ is the measure of the angle which the straight line makes with the positive direction of $\mathcal{X}$-axis

4. If $\vec{u} = (a , b)$ is a direction vector of the straight line , then its slope = $\frac{b}{a}$

5. Slope of a straight line is positive if it makes an acute angle with the positive direction of $\mathcal{X}$-axis.

6. Slope of a straight line is negative if it makes an obtuse angle with the positive direction of $\mathcal{X}$-axis.

7. Slope of $\mathcal{X}$-axis = slope of any straight line parallel to $\mathcal{X}$-axis (horizontal) = zero.

8. Slope of y-axis and slope of any straight line parallel to y-axis (vertical) is undefined

The relation between two parallel straight lines and two perpendicular straight lines :

If L_1 , L_2 are two straight lines , their slopes are m_1 , m_2 respectively, then :

1. $L_1 // L_2 \Leftrightarrow m_1 = m_2$

2. $L_1 \perp L_2 \Leftrightarrow m_1 \times m_2 = -1$ (unless one of them is parallel to one of the two coordinate axes)

Equation of the straight line :

1. Given a point on it $(\mathcal{X}_1 , y_1)$ and the slope (m) is $y - y_1 = m(\mathcal{X} - \mathcal{X}_1)$

2. Given the slope (m) and the intercepted part from y-axis (c) is $y = m\mathcal{X} + c$

3. Given the two intercepted parts from the two coordinate axes is $\frac{\mathcal{X}}{a} + \frac{y}{b} = 1$

Remarks

1. The equation of the straight line which is parallel to $\mathcal{X}$-axis and passes through the point (ℓ , k) is $y = k$
2. The equation of the straight line which is parallel to y-axis and passes through the point (ℓ , k) is $\mathcal{X} = \ell$
3. The equation of $\mathcal{X}$-axis is $y = 0$
4. The equation of y-axis is $\mathcal{X} = 0$
5. The equation of the straight line which passes through the origin point is $y = m\mathcal{X}$
6. To find the points of intersection of the curve with the $\mathcal{X}$-axis, put $y = 0$ and find values of $\mathcal{X}$
7. To find the points of intersection of the curve with the y-axis , put $\mathcal{X} = 0$ and find values of y
8. To find the points of intersection of two curves, we solve their equations simultaneously.

The equations of the tangent and normal to a curve.

If $(\mathcal{X}_1 , y_1)$ is a point lying on the curve of the function f where $y = f(\mathcal{X})$, and m is the slope of the tangent at this point.

i.e. $m = \left[\frac{dy}{d\mathcal{X}}\right]_{(\mathcal{X}_1 , y_1)}$, then :

- The equation of the tangent to the curve at the point $(\mathcal{X}_1 , y_1)$ is $y - y_1 = m(\mathcal{X} - \mathcal{X}_1)$
- The equation of the normal to the curve at the point $(\mathcal{X}_1 , y_1)$ is : $y - y_1 = \frac{-1}{m}(\mathcal{X} - \mathcal{X}_1)$

Important rules of integration

1. $\int a \, d\mathcal{X} = a\mathcal{X} + c$
2. $\int [f(\mathcal{X}) \pm g(\mathcal{X})] \, d\mathcal{X} = \int f(\mathcal{X}) \, d\mathcal{X} \pm \int g(\mathcal{X}) \, d\mathcal{X}$
3. $\int \mathcal{X}^n \, d\mathcal{X} = \frac{\mathcal{X}^{n+1}}{n+1} + c \quad , \quad n \neq -1$
4. $\int (a\mathcal{X} + b)^n \, d\mathcal{X} = \frac{(a\mathcal{X} + b)^{n+1}}{a(n+1)} + c \quad , \quad n \neq -1$
5. $\int \left(f(\mathcal{X})\right)^n \grave{f}(\mathcal{X}) \, d\mathcal{X} = \frac{\left(f(\mathcal{X})\right)^{n+1}}{n+1} + c \quad , \quad n \neq -1$

6. $* \int \sin x \, dx = -\cos x + c$ $\quad * \int \cos x \, dx = \sin x + c$

$* \int \sec^2 x \, dx = \tan x + c$ $\quad * \int \csc^2 x \, dx = -\cot x + c$

$* \int \sec x \tan x \, dx = \sec x + c$ $\quad * \int \csc x \cot x \, dx = -\csc x + c$

7. $* \int \sin (a x + b) \, dx = \frac{-1}{a} \cos (a x + b) + c$

$* \int \cos (a x + b) \, dx = \frac{1}{a} \sin (a x + b) + c$

$* \int \sec^2 (a x + b) \, dx = \frac{1}{a} \tan (a x + b) + c$

$* \int \csc^2 (a x + b) \, dx = \frac{-1}{a} \cot (a x + b) + c$

$* \int \sec (a x + b) \tan (a x + b) \, dx = \frac{1}{a} \sec (a x + b) + c$

$* \int \csc (a x + b) \cot (a x + b) \, dx = \frac{-1}{a} \csc (a x + b) + c$

Generally

1. $\int \sin \big(f(x)\big) \times f'(x) \, dx = -\cos \big(f(x)\big) + c$

2. $\int \cos \big(f(x)\big) \times f'(x) \, dx = \sin \big(f(x)\big) + c$

3. $\int \sec^2 \big(f(x)\big) \times f'(x) \, dx = \tan \big(f(x)\big) + c$

4. $\int \csc^2 \big(f(x)\big) \times f'(x) \, dx = -\cot \big(f(x)\big) + c$

5. $\int \sec \big(f(x)\big) \times \tan \big(f(x)\big) \times f'(x) \, dx = \sec \big(f(x)\big) + c$

6. $\int \csc \big(f(x)\big) \times \cot \big(f(x)\big) \times f'(x) \, dx = -\csc \big(f(x)\big) + c$

Remarks

1. $\sin^2 x = \frac{1}{2} - \frac{1}{2} \cos 2x$

2. $\cos^2 x = \frac{1}{2} + \frac{1}{2} \cos 2x$

3. $\frac{d}{dx} \int f(x) \, dx = f(x)$ while $\int \left[\frac{d}{dx} \big(f(x)\big)\right] dx = f(x) + c$

For example : $\frac{d}{dx} \int x^5 \, dx = x^5$ while $\int \left[\frac{d}{dx} (x^5)\right] dx = x^5 + c$

UNIT

ONE

Differentiation and its applications

Lesson 1 Implicit and parametric differentiation.

Lesson 2 Higher derivatives of the function.

Lesson 3 Derivatives of exponential and logarithmic functions.

Lesson 4 Related time rates.



Unit 1

Lesson One

Implicit and parametric differentiation

Implicit Differentiation

- If we have two equations : $y = x^2 + 3$, $5x^2 + 2xy + y^2 = 0$

We notice that :

The first equation : y is defined by a function of x and it is called explicit function written in the form $[y = f(x)]$

But , the second equation is not a function and is called implicit relation and written in the form $f(x, y) = 0$

The differentiation in this case is called implicit differentiation.

For example :

If $x \neq 1$, $y = \frac{x}{1-x}$ (1)

Then : y is a function of x ,

by multiplying each side by $(1 - x)$ we get :

$y(1-x) = x$ $\quad\quad$ $\therefore y - xy - x = 0$ (2)

Note that : (1) , (2) are two forms of the same function , the first form is called explicit function , and the other is called implicit function.

From the relation (1) :

$y = \frac{x}{1-x}$, then $\frac{dy}{dx} = \frac{(1-x) - x \times -1}{(1-x)^2} = \frac{1}{(1-x)^2}$

and from the relation (2) :

$y - xy - x = 0$ by derivative with respect to x (implicit differentiation)

$\therefore \frac{d}{dx}(y) - \frac{d}{dx}(xy) - \frac{d}{dx}(x) = 0$

$\therefore \frac{dy}{dx} - \left(x\frac{dy}{dx} + y\right) - 1 = 0$ $\quad$ $\therefore \frac{dy}{dx} - x\frac{dy}{dx} - y - 1 = 0$ $\quad$ $\therefore \frac{dy}{dx}(1-x) = y + 1$

$\therefore \frac{dy}{dx} = \frac{y+1}{1-x} = \frac{\frac{x}{1-x} + 1}{1-x} = \frac{x + 1 - x}{(1-x)^2} = \frac{1}{(1-x)^2}$

that is the same result which we get from the 1st form.

* Normally if we want to find $\frac{dy}{dx}$ from implicit relation , then we can convert it into explicit relation and then we use the rules of differentiation to find $\frac{dy}{dx}$

But some times we can't convert the implicit relation into explicit

(for example $x^2 - 3xy - y^3 = 4$)

because we can't separate the variable y in independent side , then we differentiate by implicit differentiation of the two sides of the relation. With respect to one of the two variables x or y to get $\frac{dy}{dx}$ or $\frac{dx}{dy}$

Remark

There exist some relations each of them is defined by two explicit functions. (not by one function)

For example :

The relation $x^2 + y^2 = 4$, is not a function because for any value of x there are two corresponding values of the variable y because $y = \pm\sqrt{4 - x^2}$

, then this relation is represented by two explicit functions.

The first : $y = \sqrt{4 - x^2}$,

its derivative is : $\frac{dy}{dx} = \frac{-2x}{2\sqrt{4-x^2}} = \frac{-x}{\sqrt{4-x^2}} = \frac{-x}{y}$

The second : $y = -\sqrt{4 - x^2}$,

its derivative is : $\frac{dy}{dx} = -\left(\frac{-2x}{2\sqrt{4-x^2}}\right) = \frac{-x}{-\sqrt{4-x^2}} = \frac{-x}{y}$

And if we differentiate the relation : $x^2 + y^2 = 4$ with respect to x we get :

$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(4)$

$\therefore 2x + 2y\frac{dy}{dx} = 0$ $\qquad$ $\therefore \frac{dy}{dx} = \frac{-x}{y}$

$\therefore$ It is the same result when we differentiate the two explicit functions.

$y = \sqrt{4 - x^2}$ and $y = -\sqrt{4 - x^2}$

From the previous we deduce that

The differentiation of the relation with respect to one of its variables (x or y) doesn't differ whether we use its explicit or implicit form.

Example 1

Find $\frac{dy}{dx}$ for the curve :

$x^2 + 4xy - 3y^2 - 2x = 3$ at the point $(3, 0)$

Solution

$x^2 + 4xy - 3y^2 - 2x = 3$ (by differentiating with respect to x)

$$\therefore 2x + 4\left(x\frac{dy}{dx} + y\right) - 3 \times 2y\frac{dy}{dx} - 2 = 0$$

$$\therefore 2x + 4x\frac{dy}{dx} + 4y - 6y\frac{dy}{dx} - 2 = 0$$

$$\therefore (4x - 6y)\frac{dy}{dx} = 2 - 2x - 4y$$

$$\therefore \frac{dy}{dx} = \frac{2 - 2x - 4y}{4x - 6y} = \frac{1 - x - 2y}{2x - 3y}$$

$$\therefore \left[\frac{dy}{dx}\right]_{(3,0)} = \frac{-1}{3}$$

Remember

$$\frac{d}{dx}(y^n) = n\,y^{n-1}\frac{dy}{dx}$$

Example 2

Find the measure of the positive angle which the tangent to the curve :

$x^2 + y^2 - 6x - 8y + 17 = 0$ makes with the positive direction of x-axis at the point $(1, 2)$

Solution

$x^2 + y^2 - 6x - 8y + 17 = 0$ (by differentiating with respect to x)

$$\therefore 2x + 2y\frac{dy}{dx} - 6 - 8\frac{dy}{dx} = 0$$

$$\therefore (2y - 8)\frac{dy}{dx} = 6 - 2x$$

$$\therefore \frac{dy}{dx} = \frac{6 - 2x}{2y - 8} = \frac{3 - x}{y - 4}$$

$$\therefore \left[\frac{dy}{dx}\right]_{(1,2)} = -1$$

$$\therefore \tan\theta = -1$$

$$\therefore m(\angle\theta) = 135°$$

Example 3

Find $\frac{dy}{dx}$ for each of the following :

1. $x^5 y^5 = 1$

2. $x^7 + y^7 = 2$

3. $y^3 + 2x^2 y - 3x y^2 = 0$

4. $\sin 3y - \cos 2x = 0$

5. $x \cos y + y \sin 2x = 1$

6. $\sec 2x + \cot y = xy$

Solution

1. $x^5 y^5 = 1$

$\therefore x \frac{dy}{dx} + y = 0$

$\therefore xy = 1$ (by differentiating with respect to x)

$\therefore \frac{dy}{dx} = \frac{-y}{x}$

2. $x^7 + y^7 = 2$ (by differentiating with respect to x)

$\therefore 7x^6 + 7y^6 \frac{dy}{dx} = 0$

$\therefore \frac{dy}{dx} = \frac{-x^6}{y^6} = -\left(\frac{x}{y}\right)^6$

3. $y^3 + 2x^2 y - 3x y^2 = 0$ (by differentiating with respect to x)

$\therefore 3y^2 \frac{dy}{dx} + 2\left(x^2 \frac{dy}{dx} + y \times 2x\right) - 3\left(x \times 2y \frac{dy}{dx} + y^2\right) = 0$

$\therefore 3y^2 \frac{dy}{dx} + 2x^2 \frac{dy}{dx} + 4xy - 6xy \frac{dy}{dx} - 3y^2 = 0$

$\therefore \frac{dy}{dx}(3y^2 + 2x^2 - 6xy) = 3y^2 - 4xy$

$\therefore \frac{dy}{dx} = \frac{3y^2 - 4xy}{3y^2 + 2x^2 - 6xy}$

4. $\sin 3y - \cos 2x = 0$ (by differentiating with respect to x)

$\therefore 3 \cos 3y \times \frac{dy}{dx} - (-2 \sin 2x) = 0$

$\therefore 3 \cos 3y \times \frac{dy}{dx} + 2 \sin 2x = 0$

$\therefore \frac{dy}{dx} = \frac{-2 \sin 2x}{3 \cos 3y}$

5. $x \cos y + y \sin 2x = 1$ (by differentiating with respect to x)

$\therefore \left[x \times \left(-\sin y \times \frac{dy}{dx}\right) + \cos y\right] + \left[y \times 2 \cos 2x + \sin 2x \times \frac{dy}{dx}\right] = 0$

$\therefore -x \sin y \times \frac{dy}{dx} + \cos y + 2y \cos 2x + \sin 2x \times \frac{dy}{dx} = 0$

$\therefore (\sin 2x - x \sin y) \frac{dy}{dx} = -2y \cos 2x - \cos y$

$\therefore \frac{dy}{dx} = \frac{-2y \cos 2x - \cos y}{\sin 2x - x \sin y}$

6. $\sec 2\mathcal{X} + \cot y = \mathcal{X}\, y$ (by differentiating with respect to $\mathcal{X}$)

$\therefore 2\sec 2\mathcal{X}\tan 2\mathcal{X} - \csc^2 y \times \frac{dy}{d\mathcal{X}} = \mathcal{X}\frac{dy}{d\mathcal{X}} + y$

$\therefore \mathcal{X}\frac{dy}{d\mathcal{X}} + \csc^2 y \times \frac{dy}{d\mathcal{X}} = 2\sec 2\mathcal{X}\tan 2\mathcal{X} - y$

$\therefore \frac{dy}{d\mathcal{X}}(\mathcal{X} + \csc^2 y) = 2\sec 2\mathcal{X}\tan 2\mathcal{X} - y \qquad \therefore \frac{dy}{d\mathcal{X}} = \frac{2\sec 2\mathcal{X}\tan 2\mathcal{X} - y}{\mathcal{X} + \csc^2 y}$

Example 4

If $\mathcal{X} = (y^5 - 1)^8$, prove that : $\frac{dy}{d\mathcal{X}} = \frac{y^5 - 1}{40\,\mathcal{X}\,y^4}$

Solution

$\because \mathcal{X} = (y^5 - 1)^8$ (by differentiating with respect to y)

$\therefore \frac{d\mathcal{X}}{dy} = 8(y^5 - 1)^7 \times 5y^4 \qquad \therefore \frac{d\mathcal{X}}{dy} = 40\,y^4(y^5 - 1)^7$

$\therefore \frac{dy}{d\mathcal{X}} = \frac{1}{40\,y^4(y^5 - 1)^7} \times \frac{y^5 - 1}{y^5 - 1} = \frac{y^5 - 1}{40\,y^4(y^5 - 1)^8} = \frac{y^5 - 1}{40\,\mathcal{X}\,y^4}$

Remark

$\frac{dy}{d\mathcal{X}} = 1 \div \frac{d\mathcal{X}}{dy}$

Example 5

If $f(\mathcal{X}) = \frac{1}{\mathcal{X} - 2}$, $g(\mathcal{X}) = \mathcal{X}^2$, find : $\frac{d}{d\mathcal{X}}[(f \circ g)(\mathcal{X})]$ at $\mathcal{X} = -1$

Solution

$\because (f \circ g)(\mathcal{X}) = f\big(g(\mathcal{X})\big) = f(\mathcal{X}^2) = \frac{1}{\mathcal{X}^2 - 2}$

$\therefore \frac{d}{d\mathcal{X}}[(f \circ g)(\mathcal{X})] = \frac{-2\mathcal{X}}{(\mathcal{X}^2 - 2)^2}$

At $\mathcal{X} = -1$: $\therefore \frac{d}{d\mathcal{X}}[(f \circ g)(\mathcal{X})] = \frac{-2(-1)}{(1 - 2)^2} = 2$

Another solution :

$\because \frac{d}{d\mathcal{X}}[(f \circ g)(\mathcal{X})] = \frac{d}{d\mathcal{X}}\big[f\big(g(\mathcal{X})\big)\big] = \grave{f}\big(g(\mathcal{X})\big) \times \grave{g}(\mathcal{X})$

, $\because \grave{f}(\mathcal{X}) = \frac{-1}{(\mathcal{X} - 2)^2}$, $\grave{g}(\mathcal{X}) = 2\mathcal{X}$

At $\mathcal{X} = -1$: $\therefore g(-1) = 1$, $\grave{g}(-1) = -2$

At $\mathcal{X} = 1$: $\therefore \grave{f}(1) = -1$

Remember

If $y = (f \circ g)(\mathcal{X}) = f\big(g(\mathcal{X})\big)$,

then $\frac{dy}{d\mathcal{X}} = \grave{f}\big(g(\mathcal{X})\big) \cdot \grave{g}(\mathcal{X})$

$\therefore \frac{d}{d\mathcal{X}} = [(f \circ g)(\mathcal{X})] = \grave{f}\big(g(-1)\big) \times \grave{g}(-1) = \grave{f}(1) \times \grave{g}(-1) = -1 \times -2 = 2$

Parametric Differentiation

- If the x-coordinate and y-coordinate of point (x, y) can be expressed as a function in a third variable t (it is called the parameter) of the two equations :

  $x = f(t)$ and $y = g(t)$ where.

  The two equations together represent an equation to one curve for all values of t that belong to the common domain of f , g expressed in the parametric form and every value of (t) get one point (x_t , y_t) and all these points form the curve.

From the previous we deduce that

- If $y = f(t)$, $x = g(t)$ are parametric form of the equation to the curve. Where f , g are two differentiable functions with respect to t

, then : $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$

Remark

We can convert the curve equation from the parametric form to the cartesian form (explicit or implicit) by canceling the parametric variable t from the equations of x , y and the new form is called cartesian form.

For example :

1. If the two parametric equations of a curve are :

$x = 2t$ (1) , $y = t^2$ (2)

, then from (1) : $t = \frac{x}{2}$

by substitute in (2) : $\therefore y = \left(\frac{x}{2}\right)^2 = \frac{x^2}{4}$

$\therefore y = \frac{1}{4} x^2$ « cartesian form ».

2. If the two parametric equations of a curve are :

$x = 3 \cos\theta$, $y = 3 \sin\theta$, then $x^2 = 9 \cos^2\theta$, $y^2 = 9 \sin^2\theta$

i.e. $x^2 + y^2 = 9 (\cos^2\theta + \sin^2\theta)$

i.e. $x^2 + y^2 = 9$ "cartesian form"

Example 6

Find $\frac{dy}{dx}$ for each of the following at the indicated value of t :

1. $x = 2at^2$, $y = at^4$ at $t = 4$

2. $x = 2\cos^3\theta$, $y = 5\sin^3\theta$ at $\theta = \frac{\pi}{4}$

Solution

1. $\because x = 2at^2$ $\therefore \frac{dx}{dt} = 4at$

, $\because y = at^4$ $\therefore \frac{dy}{dt} = 4at^3$

$\therefore \frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt} = \frac{4at^3}{4at} = t^2$ $\therefore \left[\frac{dy}{dx}\right]_{t=4} = 16$

2. $\because x = 2\cos^3\theta$ $\therefore \frac{dx}{d\theta} = 6\cos^2\theta(-\sin\theta) = -6\sin\theta\cos^2\theta$

, $\because y = 5\sin^3\theta$ $\therefore \frac{dy}{d\theta} = 15\sin^2\theta(\cos\theta) = 15\sin^2\theta\cos\theta$

$\therefore \frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta} = \frac{15\sin^2\theta\cos\theta}{-6\sin\theta\cos^2\theta} = \frac{-5\sin\theta}{2\cos\theta} = \frac{-5}{2}\tan\theta$

$\therefore \left[\frac{dy}{dx}\right]_{\theta=\frac{\pi}{4}} = \frac{-5}{2}\tan\frac{\pi}{4} = \frac{-5}{2}$

Example 7

Find : The derivative of $\sqrt{5+x^2}$ with respect to $\frac{x}{x+2}$ at $x = 2$

Solution

Put $y = \sqrt{5+x^2}$, $z = \frac{x}{x+2}$ $\therefore$ The requirement is finding $\frac{dy}{dz}$ at $x = 2$

, $\because \frac{dy}{dx} = \frac{2x}{2\sqrt{5+x^2}} = \frac{x}{\sqrt{5+x^2}}$, $\frac{dz}{dx} = \frac{x+2-x}{(x+2)^2} = \frac{2}{(x+2)^2}$

$\therefore \frac{dy}{dz} = \frac{dy}{dx} \div \frac{dz}{dx}$ $\therefore \frac{dy}{dz} = \frac{x}{\sqrt{5+x^2}} \div \frac{2}{(x+2)^2} = \frac{x(x+2)^2}{2\sqrt{5+x^2}}$

$\therefore \left(\frac{dy}{dz}\right)_{x=2} = \frac{16}{3}$

Example 8

Find the value of the parameter t at which the curve :

$x = 4t^3 - 15t^2 - 18t + 4$, $y = 2t^2 - 3t + 7$

1. Has vertical tangent.

2. Has horizontal tangent.

Solution

$\because x = 4t^3 - 15t^2 - 18t + 4$

$\therefore \frac{dx}{dt} = 12t^2 - 30t - 18$

, $\because y = 2t^2 - 3t + 7$

$\therefore \frac{dy}{dt} = 4t - 3$

$\because$ Slope of the tangent to the curve at the point (x, y) equals : $\frac{dy}{dx}$

where $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt} = \frac{4t - 3}{12t^2 - 30t - 18}$

1. In case the vertical tangent , then its slope is undefined :

$\therefore 12t^2 - 30t - 18 = 0$

$\therefore 2t^2 - 5t - 3 = 0$

$\therefore (2t + 1)(t - 3) = 0$

$\therefore t = \frac{-1}{2}$, $t = 3$

$\therefore$ The curve has vertical tangent at $t = \frac{-1}{2}$ and $t = 3$

2. In case the horizontal tangent , then its slope = 0

$\therefore 4t - 3 = 0$

$\therefore t = \frac{3}{4}$

$\therefore$ The curve has horizontal tangent at $t = \frac{3}{4}$

Notice that

* If the tangent is horizontal :
  , then the slope of tangent = 0
  i.e the numerator = 0
* If the tangent is vertical :
  , then the slope of tangent is undefined
  the denominator = 0

Example 9

Find the equations of the tangent and the normal to the curve :

$\mathcal{X} = n^2 - 2$, $y = n^4 + 3n^2 + 1$ at $n = 1$

Solution

$\because \mathcal{X} = n^2 - 2$

$\therefore \frac{d\mathcal{X}}{dn} = 2n$

, $\because y = n^4 + 3n^2 + 1$

$\therefore \frac{dy}{dn} = 4n^3 + 6n$

, $\because \frac{dy}{d\mathcal{X}} = \frac{dy}{dn} \div \frac{d\mathcal{X}}{dn}$

$\therefore \frac{dy}{d\mathcal{X}} = \frac{4n^3 + 6n}{2n} = 2n^2 + 3$

, at $n = 1$: $\frac{dy}{d\mathcal{X}} = 2(1)^2 + 3 = 5$

Remember that

At the point $(\mathcal{X}_1 , y_1)$ which lies on the curve of the function f where : $y = f(\mathcal{X})$, then :

- Slope of tangent (m) $= \left[\frac{dy}{d\mathcal{X}}\right]_{(\mathcal{X}_1 , y_1)}$
- Equation of the tangent to the curve is $y - y_1 = m(\mathcal{X} - \mathcal{X}_1)$
- Equation of the normal to the curve is $y - y_1 = \frac{-1}{m}(\mathcal{X} - \mathcal{X}_1)$

$\therefore$ The slope of the tangent $= 5$, the slope of the normal $= -\frac{1}{5}$

, $\mathcal{X} = (1)^2 - 2 = -1$, $y = (1)^4 + 3(1)^2 + 1 = 5$

i.e. The point $(-1 , 5)$ lies on the curve and

the equation of the tangent is $(y - 5) = 5(\mathcal{X} + 1)$

i.e. $y - 5\mathcal{X} - 10 = 0$

, the equation of the normal is $(y - 5) = -\frac{1}{5}(\mathcal{X} + 1)$

i.e. $\mathcal{X} + 5y - 24 = 0$